Problem statement

Problem Statement No.1: Nutrition Agent

The Challenge

Good nutrition is the foundation of health, but individuals often struggle with **diet planning, portion control, and tracking nutritional intake**. Information about healthy diets, disease-specific nutrition, and food composition is scattered across unreliable sources. Without **personalized, real-time guidance**, many fail to maintain balanced diets or make informed food choices.

The Objective

Build an **AI-powered Nutrition Agent** that delivers **personalized dietary insights and guidance** for individuals, families, and health professionals. The solution should:

1. **Personalized Diet Plans :–** Recommend meals based on age, health conditions, allergies, cultural preferences, and fitness goals.
2. **Food Data RAG Retrieval :–** Fetch and summarize nutritional values (calories, macros, vitamins) from reliable sources.
3. **Diet Tracking & Feedback :–** Allow users to log meals via text, images, or voice, and provide instant nutritional analysis.
4. **Preventive Health Focus :–** Suggest diets for chronic disease management (e.g., diabetes-friendly or heart-healthy plans).
5. **Multi-Agent System :–**
    - Nutrition Knowledge Agent (fetches nutritional data)
    - Diet Recommendation Agent (personalized meal plans)
    - Health Advisory Agent (preventive health suggestions)
    - Food Log & Feedback Agent (meal tracking + analysis)
6. Visualization Dashboard :– Show daily/weekly nutrient intake, deficiencies, and progress toward health goals.

Tech Stack (Mandatory)

- IBM Langflow / IBM Orchestrate + IBM Models

Problem Statement No.2: Agentic AI for Mental Health Awareness & Suicide Prevention

The Challenge

Mental health issues such as depression, anxiety, and suicidal thoughts are growing worldwide. Stigma, lack of awareness, and limited access to timely help prevent many individuals from seeking support. Traditional systems are often reactive, leaving gaps in early detection and preventive care.

The Objective

Build an Agentic AI-powered Mental Health Awareness & Suicide Prevention Agent that is empathetic, proactive, and safe. The solution should:

1. **Promote Awareness & Education** – Share accessible resources on mindfulness, coping strategies, and self-care.
2. **Enable Early Detection (RAG)** – Use Retrieval-Augmented Generation to identify distress signals in conversations, journaling inputs, or safe online data.
3. **Provide Empathetic Support –** Build a conversational agent that listens actively and responds with empathy (with clear disclaimers: not a substitute for professional medical advice).
4. **Predict & Prevent Risks –** Forecast stress peaks, suggest preventive strategies, and issue alerts when high-risk signals are detected (with consent).
5. **Connect to Human Support –** Link users to helplines, counselors, or community support for timely intervention.

Tech Stack (Mandatory)

- IBM Langflow / IBM Orchestrate + IBM Models

Problem Statement No.3: AI Legal Aid Multi-Agent System

The Challenge

Access to legal advice is often expensive, time-consuming, and complex. Individuals and small businesses struggle to understand contracts, rights, and compliance due to scattered resources and legal jargon. Traditional legal aid systems are limited by availability of experts and lack of real-time support.

The Objective

Build an AI-powered Multi-Agent Legal Aid System that can:

1. **Legal Information Retrieval (RAG)** – Use Retrieval-Augmented Generation to extract and summarize relevant laws, precedents, and compliance rules from large legal corpora.
2. **Document Parsing & Analysis –** Automatically analyze contracts, policies, and notices to highlight key clauses, obligations, and potential risks.
3. **Conversational Legal Assistant –** Provide empathetic, easy-to-understand explanations of legal concepts through natural dialogue (with disclaimers: *not a substitute for a lawyer*).
4. **Specialized Multi-Agents –**
    - Contract Reviewer Agent: Flag risks and inconsistencies.
    - Compliance Checker Agent: Verify adherence to rules/regulations.
    - Q&A RAG Agent: Answer user-specific queries.
    - Case Research Agent: Fetch relevant judgments or case summaries.
5. **User-Friendly Dashboard –** Deliver recommendations, case insights, and document highlights in a structured, visual format.

Tech Stack (Mandatory)

- IBM Langflow / IBM Orchestrate + IBM Models

Problem Statement No.4: Sequential Task Agent – Investment Analyst

The Challenge

Investment analysts need to process multiple layers of information—financial reports, stock performance data, market news, and analyst opinions. Much of this information is fragmented and requires a sequence of tasks (data gathering, validation, analysis, comparison, and reporting) to generate useful insights. Manual workflows are time-consuming and prone to errors. The challenge is not just to collect market data but to automate sequential tasks that lead to actionable investment recommendations.

The Objective

Build a **Sequential Task Agent** that acts as an intelligent investment analyst. The solution should:

1. **Automate Data Gathering:** Collect financial data from multiple trusted sources such as company reports, stock exchanges, and market news feeds.
2. **Perform Multi-Step Analysis:** Sequentially process data (e.g., extract KPIs → calculate financial ratios → benchmark against competitors → analyze market sentiment).
3. **Generate Predictive Insights:** Identify trends, detect risks, and provide forward-looking recommendations (e.g., *"Stock X shows declining margins—consider underweighting in portfolio"*).
4. **Deliver Structured Reports:** Provide an easy-to-understand dashboard with investment summaries, trend charts, risk indicators, and AI-generated recommendations.

Tech Stack (Mandatory)

- IBM Langflow / IBM Orchestrate + IBM Models

Problem Statement No.5: Managing Content Overload for Blog Writers

The Challenge:

Bloggers today face an overwhelming flood of scattered information—news updates, research articles, social media trends, and audience feedback. This content is fragmented, quickly outdated, and difficult to organize, making it hard for writers to focus on creating impactful blogs. The challenge is not just to collect ideas but to **synthesize, prioritize, and guide writers** toward high-quality content creation.

The Objective:

Build an **agentic AI application** that acts as an intelligent writing assistant. The solution should:

1. **Fuse Disparate Sources:** Collect real-time data from multiple channels (news sites, research papers, social media, comments) and synthesize into clear summaries with actionable blog topic suggestions.
2. **Enable Multimodal Content Curation:** Accept user inputs (text, images, or voice notes), refine ideas, and recommend blog structure, tone, or supporting visuals.
3. **Create a Predictive & Agentic Layer:** Analyze engagement patterns to predict trending topics and suggest optimal publishing time, hashtags, or angles for maximum reach.
4. **Visualize the Writing Pulse:** Provide a clean, real-time dashboard showing curated topics, sentiment analysis, and engagement forecasts to help writers align with audience interests.

Tech Stack (Mandatory):

- IBM Langflow / IBM Orchestrate + IBM Models

Problem Statement No.6: Intelligent Market Research Assistant

The Challenge

Organizations and startups often struggle with fragmented and rapidly changing market data—customer feedback, competitor moves, industry reports, and social media trends. This information is scattered across multiple sources, time-consuming to analyze, and often outdated by the time insights are generated. The challenge is not just to collect information but to **synthesize, predict, and present actionable insights** for smarter decision-making.

The Objective

Build an **agentic AI application** that acts as an intelligent market research assistant. The solution should:

1. **Fuse Disparate Sources:** Collect real-time data from news sites, analyst reports, social media, and customer reviews, and synthesize into clear, concise market snapshots.
2. **Enable Multimodal Analysis:** Accept inputs such as text, charts, or voice notes from users and refine them into structured insights, competitor comparisons, or market trends.
3. **Create a Predictive & Agentic Layer:** Analyze patterns to forecast market shifts (e.g., rising demand in a product segment) and generate alerts on competitor activities or customer sentiment changes.
4. **Visualize the Market Pulse:** Provide a dashboard with trend analysis, competitor mapping, sentiment tracking, and demand forecasts to guide data-driven business strategies.

Tech Stack (Mandatory)

- IBM Langflow / IBM Orchestrate + IBM Models

Problem Statement No.7: Research Agent

The Challenge

Researchers, students, and professionals often deal with scattered academic papers, technical articles, and experimental data spread across multiple platforms. Filtering relevant knowledge, summarizing insights, and keeping track of the latest developments is time-consuming and inefficient. The challenge is not just to collect references but to **synthesize, analyze, and generate actionable research insights** in real time.

The Objective

Build an **agentic AI application** that acts as an intelligent research companion. The solution should:

1. **Fuse Academic Sources:** Aggregate data from research papers, journals, preprints, and knowledge repositories, then synthesize them into clear, concise literature reviews.
2. **Enable Multimodal Research Input:** Accept user queries, notes, charts, or even PDFs, and convert them into structured insights, references, and summaries.
3. **Create a Predictive & Agentic Layer:** Highlight emerging trends, detect citation gaps, and predict promising directions for future research in a given domain.
4. **Visualize Research Insights:** Provide an interactive dashboard with topic clusters, citation maps, sentiment/trend analysis, and concise knowledge graphs.

Tech Stack (Mandatory)

- IBM Langflow / IBM Orchestrate + IBM Models

Problem Statement No.8: Document Q&A RAG Agent for Recipe Generator

The Challenge

Food enthusiasts and culinary professionals often rely on unstructured recipe documents, cookbooks, blogs, and user-submitted notes. Searching across these scattered resources is time-consuming, and it’s difficult to adapt recipes based on dietary needs, available ingredients, or cultural preferences. The challenge is not just to retrieve recipes but to **intelligently query, adapt, and generate personalized cooking instructions.**

The Objective

Build a **Document Q&A RAG (Retrieval-Augmented Generation) Agent** that can act as an intelligent recipe generator. The solution should:

1. **Ingest & Index Recipe Documents:** Parse cookbooks, PDFs, blogs, or uploaded recipe notes and store them in a searchable knowledge base.
2. **Enable Conversational Q&A:** Allow users to ask natural language questions (e.g., *“How can I make a sugar-free version of chocolate cake?”*) and receive precise, recipe-aware answers.
3. **Create a Personalized & Agentic Layer:** Adapt recipes based on user constraints such as available ingredients, dietary restrictions, cooking time, or cuisine preferences.
4. **Visualize Cooking Guidance:** Present results as step-by-step cooking instructions with substitution suggestions, nutritional facts, and even generate a shopping list.

Tech Stack (Mandatory)

- IBM Langflow / IBM Orchestrate + IBM Models

Problem Statement No.9 : RAG Agent for DIP Diet

The Challenge

Health-conscious individuals and nutritionists often struggle with scattered diet plans, food charts, and wellness guidelines. The **DIP Diet (Disciplined & Intelligent Plan)** approach emphasizes natural foods and lifestyle discipline, but its resources—books, blogs, and expert advice—are fragmented and hard to personalize. The challenge is not just to retrieve diet information but to **synthesize, personalize, and provide actionable dietary recommendations**.

The Objective

Build a **Vector Store RAG (Retrieval-Augmented Generation) Agent** that serves as an intelligent diet assistant for DIP followers. The solution should:

1. **Ingest & Index DIP Resources:** Collect diet plans, nutrition guides, and related health documents, and store them in a vector database for semantic search.
2. **Enable Conversational Q&A:** Allow users to ask natural questions (e.g., *"What can I eat for breakfast in the DIP diet?"*) and get clear, personalized responses.
3. **Create a Personalized & Agentic Layer**: Adapt diet suggestions based on user profile, lifestyle, medical conditions, or fitness goals while recommending food substitutions.
4. **Visualize Diet Guidance:** Provide a clean dashboard with daily/weekly meal plans, nutrition breakdown, and lifestyle tips to help users stay disciplined with DIP.

Tech Stack (Mandatory)

- IBM Langflow / IBM Orchestrate + IBM Models

Problem Statement No.10: Social Media Agent

The Challenge

Brands, influencers, and organizations struggle to keep up with the **fast-paced and fragmented world of social media**. Content trends evolve hourly, audience sentiment shifts rapidly, and competitor strategies are difficult to track. Managing multiple platforms often leads to scattered insights, missed engagement opportunities, and ineffective campaigns. The challenge is not only to collect social media data but also to **analyze, predict, and recommend actions in real time**.

The Objective

Build an **agentic AI application** that acts as an intelligent social media companion. The solution should:

1. **Fuse Multichannel Data :-** Collect and synthesize content, comments, hashtags, and competitor posts from platforms like X, LinkedIn, Instagram, and Facebook into unified insights.
2. **Enable Multimodal Interaction :-** Accept user inputs (text, images, or voice prompts) and generate creative post drafts, captions, or campaign ideas aligned with brand tone.
3. **Create a Predictive & Agentic Layer :-** Forecast trending hashtags, audience sentiment shifts, and competitor engagement patterns while suggesting the best posting schedules.
4. **Visualize Social Media Pulse :-** Provide a dashboard with real-time analytics—engagement trends, influencer mapping, sentiment tracking, and campaign performance forecasts.

Tech Stack (Mandatory)

- IBM Langflow / IBM Orchestrate + IBM Models

Problem Statement No.11: Intelligent Shopping Agent

The Challenge

Shoppers today face an overload of product options, scattered reviews, and dynamic pricing across e-commerce platforms. Finding the right product that balances price, quality, and personal preferences is time-consuming and frustrating. Traditional recommendation systems often fail to capture real-time trends, personalized needs, or contextual factors like urgency, budget, or sustainability. The challenge is not just to recommend products but to act as a smart shopping companion that helps buyers make informed and confident decisions.

The Objective

Build an agentic AI shopping assistant that provides intelligent, personalized, and real-time support. The solution should:

1. **Fuse Multisource Data:** Aggregate product details, reviews, ratings, and price comparisons across multiple e-commerce sites into unified insights.
2. **Enable Multimodal Search**: Allow users to input voice queries, images (e.g., a photo of a product), or text descriptions to find best-matching products.
3. **Create a Predictive & Agentic Layer:** Suggest purchase timing (e.g., discounts, seasonal offers), recommend alternatives, and alert users about price drops or fake reviews.
4. **Visualize Smart Shopping Pulse:** Provide a dashboard summarizing comparisons, sustainability scores, trending products, and personalized shopping lists.

Tech Stack (Mandatory)

- IBM Langflow / IBM Orchestrate + IBM Models

Problem Statement No.12: Smart Business Idea Generator Agent

The Challenge

In the digital era, innovation is crucial, but individuals, students, researchers, and organizations often struggle with idea generation. Brainstorming sessions are unstructured, inspiration is scattered across articles, videos, and social media, and existing tools focus on text-only prompts without context-awareness. The challenge is not just to suggest random ideas but to synthesize creativity, context, and relevance into intelligent, actionable ideas for diverse domains such as startups, research, content creation, or product development.

The Objective

Build an agentic AI Smart Idea Generator that empowers users with contextual, innovative, and multimodal ideation. The solution should:

1. **Fuse Knowledge Sources:** Pull insights from articles, research papers, trend analysis, and social platforms to generate contextual and current ideas.
2. **Enable Multimodal Brainstorming**: Accept inputs like text prompts, images (e.g., product sketches), or voice notes and refine them into structured, creative suggestions.
3. **Create a Predictive & Agentic Layer:** Identify trending domains (e.g., AI in healthcare, sustainable energy), forecast idea relevance, and recommend high-impact directions.
4. **Visualize Idea Maps:** Provide an interactive dashboard that organizes ideas into categories, impact scores, and feasibility maps, helping users explore and refine concepts quickly.

Tech Stack (Mandatory)

- IBM Langflow / IBM Orchestrate + IBM Models

Problem Statement No.13: Smart Study Generator Agent

The Challenge

Students today face an overwhelming flood of information—lecture notes, textbooks, research papers, YouTube tutorials, and online resources. This content is scattered, unorganized, and often not personalized to their learning pace or style. As a result, students struggle with finding the right study material, summarizing key concepts, and staying exam-ready. Traditional study tools only provide static notes or flashcards, failing to adapt dynamically to student needs. The challenge is to move beyond static resources toward an intelligent, adaptive study companion.

The Objective

Build an agentic AI-powered Smart Study Generator that personalizes and streamlines learning for students. The solution should:

1. **Fuse Learning Resources:** Aggregate content from class notes, textbooks, research articles, and online tutorials into structured, easy-to-digest study material.

2. **Enable Multimodal Input:** Allow students to upload notes, voice queries, or images (e.g., textbook pages), and generate concise summaries, flashcards, or concept maps.

3. **Create a Predictive & Agentic Layer:** Recommend personalized study plans, highlight important topics based on syllabus or past exams, and provide predictive alerts (e.g., "Focus on Topic X—frequently asked in exams").

4. **Visualize Study Progress:** Offer a real-time dashboard showing learning milestones, weak areas, practice quizzes, and performance forecasts.

Tech Stack (Mandatory)

- IBM Langflow / IBM Orchestrate + IBM Models

Problem Statement No.14: AI Agent for Smart Farming Advice

The Challenge

Farmers often struggle with low crop yield, unpredictable weather, pest infestations, and limited access to timely expert advice. Traditional farming advisory services are fragmented, slow, and not personalized to the farmer's specific land, crop, or climate conditions. With the rise of climate change, data-driven smart farming solutions are needed to empower farmers with actionable insights.

The Objective

Develop an AI-powered Smart Farming Agent that provides personalized, real-time, and sustainable agricultural advice. The solution should:

1. **Agricultural Knowledge Retrieval (RAG) –** Use Retrieval-Augmented Generation to fetch best practices, crop-specific guidelines, and government advisory updates.
2. **Weather & Soil Insights** – Integrate real-time weather forecasts, soil health data, and irrigation needs to optimize farm planning.
3. **Pest & Disease Detection –** Enable multimodal support where farmers upload crop images for early pest/disease identification and get instant recommendations.
4. **Personalized Advisory Agent –** Suggest seeds, fertilizers, irrigation schedules, and harvesting timelines based on farmer profile and land conditions.
5. **Market & Cost Optimization –** Forecast demand, suggest market prices, and recommend cost-efficient practices for better income.
6. **Multi-Agent Collaboration –**
   - Crop Advisory Agent
   - Weather & Irrigation Agent
   - Pest/Disease Detection Agent
   - Market Insights Agent

Tech Stack (Mandatory)

- IBM Langflow / IBM Orchestrate + IBM Models

Problem Statement No.15: The Daily Air Quality Action Agent

The Challenge

Air pollution is a **major public health crisis**, contributing to respiratory issues, climate change, and reduced quality of life. Citizens often receive air quality index (AQI) data, but it is presented in **technical formats**, making it difficult to understand or translate into **daily lifestyle actions**. Traditional monitoring systems lack **personalized recommendations** and proactive alerts.

The Objective

Build an **AI-powered Daily Air Quality Action Agent** that translates **air quality data into actionable insights** for citizens, policymakers, and organizations. The solution should:

1. **AQI Data Retrieval (RAG) –** Collect and interpret real-time AQI from public APIs, satellite data, or government sources.
2. **Personalized Health Recommendations** – Provide daily actionable advice (e.g., avoid outdoor exercise, use masks, keep children indoors) tailored to age, location, and health conditions.
3. **Predictive Alerts –** Forecast poor air quality days, issue early warnings, and suggest preventive measures like adjusting commute or using air purifiers.
4. **Policy & Awareness Insights –** Suggest community-level actions (e.g., carpooling, waste reduction) and summarize policy advisories for public awareness.
5. **Multi-Agent System** -
    - AQI Data Agent (fetches & interprets data)
    - Health Advisory Agent (personalized lifestyle suggestions)
    - Forecasting Agent (predicts trends & risks)
    - Community Action Agent (drives awareness campaigns)
6. **Visualization Dashboard** – Show daily AQI levels, risk alerts, and recommended actions in a clear and interactive format.

Tech Stack (Mandatory)

- IBM Langflow / IBM Orchestrate + IBM Models

Electrical Engineering:

Problem Statement No.16 – Smart Home Energy Advisor Agent

The Challenge

Rising electricity demand and energy costs make it difficult for households to understand how their daily habits impact power consumption. Most users only receive monthly bills, which provide limited insight into appliance-level energy usage. As a result, people often waste electricity unknowingly, highlighting the need for an AI-powered system that analyzes usage patterns and provides actionable energy-saving recommendations.

The Objective

Build an AI-powered Smart Home Energy Advisor Agent that analyzes household electricity consumption data and provides intelligent recommendations to optimize energy usage.

1. **Energy Consumption Analysis** – Analyze household electricity usage patterns from smart meters or appliance usage logs.
2. **Personalized Energy Recommendations** – Suggest energy-saving actions based on the user's consumption behavior and daily routines.
3. **Appliance Usage Insights** – Identify appliances that consume the most electricity and recommend optimal usage strategies.
4. **Electricity Bill Estimation** – Predict monthly electricity bills based on current usage patterns and tariff structures.
5. **Energy Efficiency Suggestions** – Recommend simple practices such as reducing peak-hour usage, scheduling heavy appliances, or upgrading to energy-efficient devices.

Tech Stack (Mandatory)

- IBM Langflow / IBM Orchestrate + IBM Models

Problem Statement No.17 - AI-Based Smart Load Balancing Agent for Buildings.

The Challenge

Modern buildings consume large amounts of electricity due to HVAC systems, lighting, elevators, and other electrical equipment. However, energy loads are often unevenly distributed, leading to peak demand spikes, inefficient power utilization, and higher electricity costs. Buildings lack intelligent systems that can dynamically monitor and balance electrical loads to ensure efficient energy usage.

The Objective

Build an **AI-Based Smart Load Balancing Agent for Buildings** that analyzes energy consumption patterns and intelligently distributes electrical loads across building systems to optimize energy usage and reduce peak demand. The solution should:

1. **Load Monitoring and Analysis –** Analyze electricity usage data from building systems such as HVAC, lighting, elevators, and other appliances.
2. **Peak Load Detection –** Identify peak electricity demand periods and detect abnormal load spikes that may cause inefficiencies or overloads.
3. **Smart Load Balancing Recommendations –** Provide intelligent recommendations to shift or distribute loads across different time periods to maintain optimal energy usage.
4. **Energy Optimization Insights –** Suggest strategies to reduce peak demand charges and improve overall building energy efficiency.

Tech Stack (Mandatory)

- IBM Langflow / IBM Orchestrate + IBM Models

Problem Statement No.18 - AI Agent for Electricity Bill Fraud Detection

The Challenge

Electricity fraud and billing anomalies lead to major financial losses for power distribution companies. With thousands of consumers and massive billing data, detecting suspicious consumption patterns manually is extremely difficult. Utilities need an intelligent AI system that can automatically analyze electricity usage data and identify potential fraud cases.

The Objective

Build an **AI Agent for Electricity Bill Fraud Detection** that analyzes electricity consumption and billing records to detect abnormal patterns and flag potential fraud cases.

The solution should:

1. **Electricity Consumption Analysis –** Analyze consumer electricity usage data to understand normal consumption behavior and trends.
2. **Fraud Pattern Identification –** Detect suspicious activities such as sudden drops in usage, unusual spikes, or inconsistencies between meter readings and billed units.
3. **AI-Based Anomaly Detection –** Automatically identify consumers whose usage patterns significantly deviate from normal historical behavior.
4. **Explainable AI Insights –** Provide clear explanations for why a particular consumer or electricity bill is flagged as suspicious.

Tech Stack (Mandatory)

- IBM Langflow / IBM Orchestrate + IBM Models

Problem Statement No 19- Electric Vehicle Charging Optimization Agent

The Challenge

With the rapid growth of electric vehicles (EVs), charging stations often face high demand, uneven usage, and inefficient energy distribution. Peak-hour charging can overload the grid and increase electricity costs for both operators and users. There is a need for an intelligent system that can analyze charging patterns and optimize EV charging schedules.

The Objective

Build an **Electric Vehicle Charging Optimization Agent** that analyzes charging station usage data and provides intelligent recommendations to optimize charging schedules and energy usage.

1. **Charging Pattern Analysis** – Analyze historical EV charging data to understand peak usage hours, charging demand, and station utilization.
2. **Charging Schedule Optimization** – Recommend optimal charging schedules to balance demand and reduce peak-hour congestion.
3. **Energy Demand Prediction** – Predict future charging demand based on historical patterns and user behavior.
4. **Cost Optimization Insights** – Suggest charging during low-tariff periods to reduce electricity costs for EV users and charging station operators.

Tech Stack (Mandatory)

- IBM Langflow / IBM Orchestrate + IBM Models

Problem Statement No 20 - AI Agent for Electricity Consumption Awareness

The Challenge

Many consumers lack awareness about how daily activities and appliance usage impact their electricity consumption. This leads to inefficient usage and unnecessary energy wastage.

The challenge is to build an AI Agent that educates users about electricity consumption patterns and promotes energy-efficient behavior.

The Objective

1. **Energy Consumption Analysis :-The** AI agent should analyze household electricity usage patterns and identify areas where energy consumption is unusually high.

2. **User Awareness and Education :-** The system should provide simple explanations that help users understand how their daily activities affect electricity consumption.

3. **Energy Saving Suggestions** :- The AI agent should provide practical tips and recommendations that help users reduce electricity usage and adopt energy-efficient practices.

4. **Conversational Interaction :-** Users should be able to ask questions related to electricity usage and receive helpful insights and suggestions from the system.

Tech Stack (Mandatory)

- IBM Langflow / IBM Orchestrate + IBM Models

Electronics and Telecommunications Engineering: (Agentic AI project)

Problem Statement No.21 : Audio Signal Processing Assistant Agent

The Challenge

Audio signal processing circuits such as amplifiers, filters, and preamplifiers are widely used in communication systems, broadcasting equipment, and consumer electronics. Engineers and students frequently encounter issues like noise, distortion, grounding problems, and incorrect filter configurations while designing or troubleshooting audio circuits.

Understanding these problems requires deep knowledge of signal processing concepts and circuit behaviour. Most users must search through multiple technical manuals, textbooks, and online forums, which can be time-consuming and inefficient.

The challenge is to build an intelligent system that can **assist users in understanding audio signal issues, diagnosing circuit problems, and suggesting design improvements**.

The Objective

Build an **AI-powered Audio Signal Processing Assistant Agent** that helps engineers and students troubleshoot and design audio signal processing circuits. The solution should.

1. **Intelligent Query Understanding :-** Allow users to ask questions about audio circuits, signal processing concepts, and troubleshooting issues using natural language.
2. **Troubleshooting Support :-** Analyze possible causes of issues such as noise, distortion, signal attenuation, or grounding problems and provide step-by-step debugging guidance.
3. **Circuit Design Recommendations :-** Suggest improvements for filters, amplifiers, and signal conditioning circuits to enhance performance.
4. **Knowledge Retrieval :-** Retrieve relevant technical knowledge from electronics and signal processing documents to support explanations and recommendations.

Tech Stack

- IBM Langflow / IBM Orchestrate + IBM Models

Problem Statement No.22: Smart IoT Device Troubleshooting Chatbot

The Challenge

IoT devices are widely used in smart homes, industries, and connected environments. However, users frequently encounter problems such as connectivity failures, configuration errors, firmware issues, and device compatibility challenges.

Troubleshooting these problems often requires technical knowledge and the ability to interpret complex documentation. Many users depend on manuals, forums, or customer support, which can delay issue resolution.

The challenge is to develop an intelligent assistant that can **quickly diagnose IoT device problems and guide users through troubleshooting steps**.

The Objective

Build a **conversational AI chatbot** that helps users identify and resolve IoT device issues through natural language interaction. The solution should:

1. **Understand Device Issues :-** Interpret user descriptions of IoT problems such as network connectivity failures, configuration errors, or device malfunctions.
2. **Retrieve Relevant Knowledge :-** Access IoT documentation and troubleshooting guides using a Retrieval-Augmented Generation (RAG) approach.
3. **Provide Step-by-Step Solutions :-** Generate clear instructions for resolving connectivity, configuration, or firmware issues.
4. **Conversational Support :-** Offer an interactive chatbot interface where users can ask questions and receive guidance in real time.

Tech Stack

- IBM Langflow / IBM Orchestrate + IBM Models

Problem Statement No.23 : AI-Based 5G Network Optimization Advisor

The Challenge

5G networks support high-speed communication, low latency, and large-scale device connectivity. However, maintaining optimal performance is challenging due to fluctuating traffic loads, spectrum congestion, and energy consumption issues.

Telecom engineers must continuously analyze network data to identify inefficiencies and optimize resource allocation. Traditional monitoring tools provide raw data but lack intelligent insights.

The challenge is to create a system that can **analyze network performance data and recommend optimization strategies automatically**.

The Objective

Build an **AI-powered 5G Network Optimization Advisor** that analyzes network traffic patterns and provides recommendations for improving network efficiency. The solution should:

1. **Analyze Network Traffic** :- Process performance metrics such as bandwidth usage, latency, and network congestion.
2. **Detect Performance Issues :-** Identify inefficient resource utilization and potential bottlenecks in the network.
3. **Provide Optimization Recommendations** :- Suggest strategies to improve bandwidth allocation, reduce congestion, and enhance energy efficiency.
4. **Visualize Network Insights :-** Display network analytics and optimization suggestions through an interactive dashboard.

Tech Stack

- IBM Langflow / IBM Orchestrate + IBM Models

Problem Statement No.24 : Smart Satellite Communication Knowledge Assistant

The Challenge

Satellite communication systems involve complex concepts such as frequency bands, signal modulation, link budgets, and communication protocols. Engineers and students often struggle to understand these topics because the knowledge is distributed across textbooks, research papers, and technical documentation.

Finding relevant information and troubleshooting communication issues can take significant time.

The challenge is to build an intelligent assistant that can **retrieve knowledge and explain satellite communication concepts in an easy and interactive manner**.

The Objective

Build a **Satellite Communication Knowledge Assistant** that helps users understand and troubleshoot satellite communication systems. The solution should:

1. **Knowledge Retrieval :-** Extract information from satellite communication documents and technical materials.
2. **Concept Explanation** :- Explain complex topics such as modulation techniques, frequency bands, and link budgets in simple language.
3. **Troubleshooting Support** :- Provide guidance for solving satellite communication problems.
4. **Conversational Interface :-** Allow users to ask questions and receive responses through an interactive chat interface.

Tech Stack

- IBM Langflow / IBM Orchestrate + IBM Models

Problem Statement No.25 : AI-Powered Signal Quality Analyzer

The Challenge

Reliable telecommunication systems depend on signal quality parameters such as Signal-to-Noise Ratio (SNR), Bit Error Rate (BER), and latency. Engineers must analyze these metrics to detect signal degradation and communication failures.

Manual analysis of signal measurement data can be complex and time-consuming, especially in large communication networks.

The challenge is to develop an intelligent system that can **analyze signal performance data and provide actionable insights for improving communication quality**.

The Objective

Build an **AI-powered Signal Quality Analyzer** that evaluates signal performance and provides recommendations to improve network reliability. The solution should:

1. **Process Signal Data :-** Analyze signal measurement datasets containing parameters such as SNR, BER, and latency.
2. **Detect Signal Issues :-** Identify anomalies, interference patterns, and signal degradation.
3. **Generate Intelligent Insights :-** Use AI to explain the causes of signal problems and recommend improvements.
4. **Predict Potential Failures :-** Forecast possible signal quality issues using historical data analysis.

Tech Stack

- IBM Langflow / IBM Orchestrate + IBM Models

Problem Statement No.26 : AI-Powered Telecom Training Assistant

The Challenge

Telecommunication technologies include complex topics such as wireless communication, network architectures, protocols, and signal processing. Students and engineers often struggle to understand these topics through traditional training methods.

Most training programs rely on static materials such as lectures, textbooks, and recorded courses, which do not provide personalized learning or interactive support.

The challenge is to build an intelligent assistant that can **support telecom education through interactive learning and instant doubt resolution**.

The Objective

Build an **AI-powered Telecom Training Assistant** that helps learners understand telecom technologies in an interactive way. The solution should:

1. **Retrieve Learning Materials :-** Access telecom training resources and technical documentation using RAG.
2. **Explain Concepts :-** Provide simplified explanations of complex telecom topics.
3. **Generate Quizzes :-** Create practice questions and quizzes to test learners' understanding.
4. **Interactive Learning :-** Allow users to ask questions and receive answers in real time through a chatbot interface.

Tech Stack

- IBM Langflow / IBM Orchestrate + IBM Models

Problem Statement No.27 : AI-Based RF Interference Detection Assistant

The Challenge

Radio Frequency (RF) interference can significantly affect communication systems such as WiFi networks, satellite communication, and mobile networks. Interference from electronic devices, overlapping frequencies, and environmental conditions can reduce signal quality and disrupt communication.

Identifying the source of RF interference requires specialized equipment and technical expertise. Engineers must manually analyze signal patterns, which can be complex and time-consuming.

The challenge is to build an intelligent system that can **detect interference patterns and recommend solutions to improve communication performance**.

The Objective

Build an **AI-Based RF Interference Detection Assistant** that analyzes RF signal data and identifies potential interference sources. The solution should:

1. **Analyze RF Signal Data :-** Process RF measurement datasets to detect abnormal signal patterns.
2. **Identify Interference Sources:-** Recognize potential causes of interference in communication systems.
3. **Provide Mitigation Strategies :-** Recommend techniques to reduce interference and improve signal quality.
4. **Visualize Signal Insights :-** Display signal analysis results and alerts through an interactive dashboard.

Tech Stack

- IBM Langflow / IBM Orchestrate + IBM Models

Problem Statement No.28: The AI Sports Match Predictor Agent

The Challenge

Sports fans, analysts, and teams constantly try to predict match outcomes, but predictions are often based on intuition, limited statistics, or subjective opinions. With massive amounts of historical match data, player statistics, and team performance metrics available today, manually analyzing all these variables becomes extremely difficult. Traditional prediction models often fail to capture deeper contextual insights such as recent player form, head-to-head performance, or strategic matchups.

The Objective

Build an AI-powered Sports Match Predictor Agent that analyzes historical match data, team statistics, and player performance to generate intelligent predictions and insights about upcoming matches.

1. **Match Data Retrieval (RAG) :–** Collect historical match statistics, team records, and player performance data from sports datasets and APIs.
2. **Performance Analysis** :– Analyze trends such as team form, player consistency, win-loss patterns, and venue performance.
3. **Match Outcome Prediction :–** Predict the likely winner or probability of outcomes using AI reasoning and statistical insights.
4. **Explainable Insights** :– Provide clear explanations behind predictions such as team form, head-to-head advantage, and key player impact.
5. **Multi-Agent System**
   - Data Collection Agent – Retrieves historical match data and player statistics.
   - Performance Analysis Agent – Evaluates player and team performance metrics.
   - Prediction Agent – Generates outcome predictions using AI reasoning.
   - Insight Generation Agent – Converts predictions into simple explanations for fans and analysts.
6. **Visualization Dashboard**
   Display match predictions, team performance trends, win probabilities, and key influencing factors in an interactive dashboard.

Tech Stack

- IBM Langflow / IBM Orchestrate + IBM Models

Problem Statement No.29: The AI Game Strategy Advisor Agent

The Challenge

Modern sports teams rely heavily on strategy and tactical planning. However, analyzing past matches, opponent strategies, and player performance patterns requires extensive manual work by analysts and coaching staff. Many teams lack tools that can quickly analyze historical gameplay data and generate strategic insights.

The Objective

Build an AI-powered Game Strategy Advisor Agent that analyzes previous match data, team formations, and opponent strategies to recommend optimized game plans.

1. **Match Data Analysis (RAG)** :– Retrieve historical match data, formations, and tactical patterns.
2. **Opponent Strategy Analysis :–** Detect patterns in opponent play styles, strengths, and weaknesses.
3. **Tactical Recommendation Engine :–** Suggest optimal strategies such as formations, defensive setups, or attacking tactics.
4. **Real-Time Strategic Insights –** Provide quick recommendations before or during matches.
5. **Multi-Agent System**
    - Match Analysis Agent – Analyzes historical gameplay data.
    - Opponent Strategy Agent – Identifies tactical patterns used by opposing teams.
    - Strategy Recommendation Agent – Suggests optimal game plans.
    - Insight Explanation Agent – Converts complex analysis into actionable coaching advice.
6. **Visualization Dashboard**
   Show team formations, tactical comparisons, opponent weaknesses, and recommended strategies in a clear visual interface.

Tech Stack

- IBM Langflow / IBM Orchestrate + IBM Models

Problem Statement No.30: The AI Sports Highlight Generator Agent

The Challenge

Sports fans and media platforms constantly look for highlights from matches such as goals, wickets, decisive plays, and turning points. Currently, identifying and editing these highlights requires manual video analysis, which is time-consuming and resource intensive.

The Objective

Build an AI-powered Sports Highlight Generator Agent that automatically detects key match events and generates highlight summaries.

1. **Match Event Data Processing :–** Collect and interpret event data such as goals, wickets, fouls, or match-changing plays.
2. **Key Moment Detection :–** Identify turning points and exciting moments in matches.
3. **Automated Highlight Generation :–** Generate short highlight summaries or timestamps.
4. **Fan-Friendly Summaries :–** Produce engaging highlight descriptions for fans and media platforms.
5. **Multi-Agent System**
   - Event Detection Agent – Processes match event data and identifies important events.
   - Highlight Identification Agent – Detects turning points and exciting moments.
   - Content Generation Agent – Creates highlight summaries and descriptions.
   - Media Output Agent – Formats highlights for sports platforms and fan engagement.
6. **Visualization Dashboard**
   Display highlight timelines, key match moments, and automatically generated summaries for quick viewing.

Tech Stack

- IBM Langflow / IBM Orchestrate + IBM Models

Problem Statement No.31: The Player Recruitment Intelligence Agent

The Challenge

Sports teams constantly scout for new talent, but evaluating players requires analyzing massive amounts of performance data, scouting reports, and historical match statistics. Traditional scouting methods rely heavily on subjective observations and manual analysis.

The Objective

Build an AI-powered Player Recruitment Intelligence Agent that analyzes player statistics, performance data, and scouting reports to recommend suitable players for teams.

1. **Player Data Retrieval (RAG) :–** Collect player statistics, match records, and scouting reports.
2. **Performance Evaluation :–** Analyze metrics such as scoring ability, defensive contributions, and consistency.
3. **Team Fit Analysis :–** Match player skills with team strategy and requirements. Recruitment Recommendation Engine – Generate ranked player recommendations.
4. **Multi-Agent System**
   - Player Data Agent – Retrieves player performance statistics.
   - Performance Analysis Agent – Evaluates player capabilities and strengths.
   - Recruitment Recommendation Agent – Suggests best-fit players for teams.
   - Scouting Insight Agent – Generates recruitment reports.
5. **Visualization Dashboard**
   Display player rankings, performance comparisons, and recruitment recommendations in an interactive interface.

Tech Stack

- IBM Langflow / IBM Orchestrate + IBM Models

Problem Statement No.32: The Athlete Performance Analysis Assistant

The Challenge

Athletes generate large amounts of performance data during training and competitions, including fitness metrics, match statistics, and workload indicators. Coaches often struggle to convert this raw data into actionable insights that improve performance.

The Objective

Build an AI-powered Athlete Performance Analysis Assistant that evaluates athlete performance using training data, fitness metrics, and match statistics.

1. **Performance Data Retrieval (RAG) :–** Collect athlete statistics, training records, and fitness data.
2. **Performance Trend Analysis :–** Identify strengths, weaknesses, and improvement opportunities.
3. **Training Recommendations :-** Suggest personalized training strategies based on performance patterns.
4. **Coaching Insights :–** Provide clear summaries and reports for coaching staff.
5. **Multi-Agent System**
   - Athlete Data Agent – Retrieves training and performance data.
   - Performance Analysis Agent – Evaluates performance metrics and trends.
   - Training Recommendation Agent – Suggests improvement strategies.
   - Coaching Insight Agent – Generates performance reports.
6. **Visualization Dashboard**
   Display performance trends, fitness metrics, improvement suggestions, and athlete comparisons.

Tech Stack

- IBM Langflow / IBM Orchestrate + IBM Models

Mechanical Domain:

Problem Statement No.33: The AI Mechanical Design Optimization Agent

The Challenge
Mechanical design engineers must evaluate multiple parameters such as strength, material properties, weight, durability, and manufacturing feasibility. Traditional design optimization requires repeated simulations and manual iteration, which consumes significant engineering time and resources. Engineers often need to analyze multiple design alternatives before selecting the optimal configuration.

The Objective

Build an AI-powered Mechanical Design Optimization Agent that analyzes engineering design parameters and suggests optimized mechanical configurations.

1. **Design Data Retrieval (RAG) :–** Collect CAD parameters, design specifications, and material properties.
2. **Performance Analysis :–** Evaluate mechanical strength, stress distribution, and efficiency.
3. **Design Optimization :–** Suggest improved design configurations.
   Engineering Insights – Provide explanations behind optimization recommendations.
4. **Multi-Agent System**
   - Design Data Agent – Retrieves engineering design parameters.
   - Simulation Analysis Agent – Evaluates performance metrics.
   - Optimization Agent – Suggests improved design alternatives.
   - Engineering Insight Agent – Explains optimization decisions.
5. **Visualization Dashboard**
   Display design comparisons, optimized parameters, and predicted performance improvements.

Tech Stack

- IBM Langflow / IBM Orchestrate + IBM Models

Problem Statement No.34: The Smart Mechanical Fault Diagnosis Agent

The Challenge

Mechanical systems such as engines, pumps, and gearboxes often experience faults that are difficult to diagnose quickly. Engineers and technicians typically rely on manual inspections, experience, and trial-and-error troubleshooting methods. This process can lead to extended downtime, higher maintenance costs, and inefficient repair processes.

The Objective

Build an AI-powered Mechanical Fault Diagnosis Agent that analyzes machine symptoms, sensor data, and historical maintenance logs to identify possible mechanical faults.

1. **Fault Data Retrieval (RAG)** :– Collect historical fault records and maintenance logs.
2. Symptom Analysis :– Analyze vibration patterns, noise signals, and temperature anomalies.
3. **Root Cause Identification :–** Identify the most probable cause of failure.
   Repair Guidance – Provide recommended repair procedures.
4. **Multi-Agent System**
   - Symptom Detection Agent – Collects machine symptoms and sensor signals.
   - Fault Analysis Agent – Matches symptoms with known fault patterns.
   - Root Cause Agent – Determines the probable mechanical issue.
   - Repair Guidance Agent – Suggests troubleshooting and repair procedures.
5. **Visualization Dashboard**
   Display fault alerts, root cause explanations, and repair recommendations.

Tech Stack

- IBM Langflow / IBM Orchestrate + IBM Models

Problem Statement No.35: The AI Mechanical Safety Compliance Advisor

The Challenge

Industrial machines must comply with strict safety regulations and operational standards. Monitoring compliance across large industrial facilities can be difficult, and violations may lead to safety hazards, accidents, and regulatory penalties. Organizations require intelligent systems that can continuously monitor operations and highlight potential safety risks.

The Objective

Build an AI-powered Mechanical Safety Compliance Advisor that analyzes machine operations and safety regulations to detect compliance risks.

1. **Safety Data Retrieval (RAG) :–** Collect machine safety standards and operational logs.
2. **Risk Detection :–** Identify unsafe operating conditions.
3. **Compliance Verification :–** Compare operational parameters with safety standards.
4. **Safety Recommendations :–** Suggest corrective actions to ensure safe operation.
5. **Multi-Agent System**
    - Safety Data Agent – Collects safety standards and operational data.
    - Risk Detection Agent – Identifies hazardous conditions.
    - Compliance Agent – Verifies adherence to safety regulations.
    - Safety Recommendation Agent – Suggests corrective safety actions.
6. **Visualization Dashboard**
   Display safety alerts, compliance reports, and recommended safety improvements.

Tech Stack

- IBM Langflow / IBM Orchestrate + IBM Models

Problem Statement No.36: The AI Energy Efficiency Optimization Agent for Mechanical Systems

The Challenge

Mechanical systems such as HVAC units, compressors, turbines, and industrial machines consume large amounts of energy. Inefficient operation increases operational costs and environmental impact. Organizations often lack advanced tools to analyze energy consumption patterns and identify opportunities for optimization.

The Objective

Build an AI-powered Energy Efficiency Optimization Agent that analyzes mechanical system data and recommends improvements to reduce energy consumption.

1. **Energy Data Retrieval (RAG) :–** Collect energy consumption data and operational parameters.
2. **Efficiency Analysis :–** Identify inefficiencies in machine operations.
   Optimization Recommendations :– Suggest operational adjustments or system improvements.
3. **Energy Forecasting :–** Predict future energy consumption trends.
4. **Multi-Agent System**
   - Energy Monitoring Agent – Collects energy consumption data.
   - Efficiency Analysis Agent – Detects inefficient operations.
   - Optimization Agent – Recommends energy-saving strategies.
   - Forecasting Agent – Predicts future energy usage.
5. **Visualization Dashboard**
   Display energy consumption trends, efficiency indicators, and optimization recommendations.

Tech Stack

- IBM Langflow / IBM Orchestrate + IBM Models

Problem Statement No.37: The Manufacturing Process Quality Control Agent

The Challenge

In modern manufacturing industries, maintaining consistent product quality is critical for customer satisfaction, safety, and regulatory compliance. However, manufacturing processes often involve multiple machines, materials, and environmental conditions that can affect product quality. Small variations in temperature, pressure, machine calibration, or material properties can lead to defects such as dimensional inaccuracies, surface imperfections, or structural weaknesses.

Traditional quality control methods rely heavily on manual inspection or periodic sampling, which may fail to detect defects early in the production process. As production volumes increase, manual inspection becomes inefficient and prone to human error. Manufacturers require intelligent systems that can continuously monitor production parameters, detect anomalies, and predict quality issues before defective products are produced.

The Objective

Build an **AI-powered Manufacturing Process Quality Control Agent** that continuously monitors production parameters and identifies potential quality issues in real time.

1. **Process Data Retrieval (RAG) :–** Collect manufacturing data such as machine parameters, sensor readings, material properties, and inspection records.
2. **Defect Detection :–** Identify abnormal patterns in manufacturing processes that could lead to defects.
3. **Quality Prediction :–** Predict potential product defects before they occur.
4. **Process Optimization Recommendations :–** Suggest adjustments to machine parameters or production conditions to maintain quality standards.
5. **Automated Quality Insights :–** Generate clear reports for engineers and production managers.
6. **Multi-Agent System**
    - Process Monitoring Agent – Collects real-time data from machines and sensors in the manufacturing process.
    - Quality Analysis Agent – Analyzes production data to identify anomalies or quality deviations.
    - Defect Prediction Agent – Predicts potential defects based on historical and real-time data.
    - Process Optimization Agent – Suggests corrective actions to improve product quality.
7. **Visualization Dashboard :-**Display real-time production metrics, defect alerts, quality trends, and recommended corrective actions in a clear and interactive dashboard for production engineers.

Tech Stack (Mandatory)

Use any one of the following approaches:

- IBM Langflow / IBM Orchestrate + IBM Models